gives apparently inaccurate and seemingly ridiculous conclusions when applied toward what we call the past. Glasses of water with partially melted ice cubes do not usually start out as glasses of water with no ice cubes in which molecules of water coalesce and cool into chunks of ice, only to start melting once again. Unbound pages of *War and Peace* do not usually start thoroughly out of numerical order and through subsequent tosses get less jumbled, only to start getting more jumbled again. And going back to the kitchen, eggs do not generally start out splattered, and then coalesce into a pristine whole egg, only to splatter some time later.

Or do they?

Following the Math

Centuries of scientific investigations have shown that mathematics provides a powerful and incisive language for analyzing the universe. Indeed, the history of modern science is replete with examples in which the math made predictions that seemed counter to both intuition and experience (that the universe contains black holes, that the universe has anti-matter, that distant particles can be entangled, and so on) but which experiments and observations were ultimately able to confirm. Such developments have impressed themselves profoundly on the culture of theoretical physics. Physicists have come to realize that mathematics, when used with sufficient care, is a proven pathway to truth.

So, when a mathematical analysis of nature's laws shows that entropy should be higher toward the future *and* toward the past of any given moment, physicists don't dismiss it out of hand. Instead, something akin to a physicists' Hippocratic oath impels researchers to maintain a deep and healthy skepticism of the apparent truths of human experience and, with the same skeptical attitude, diligently follow the math and see where it leads. Only then can we properly assess and interpret any remaining mismatch between physical law and common sense.

Toward this end, imagine it's 10:30 p.m. and for the past half hour you've been staring at a glass of ice water (it's a slow night at the bar), watching the cubes slowly melt into small, misshapen forms. You have